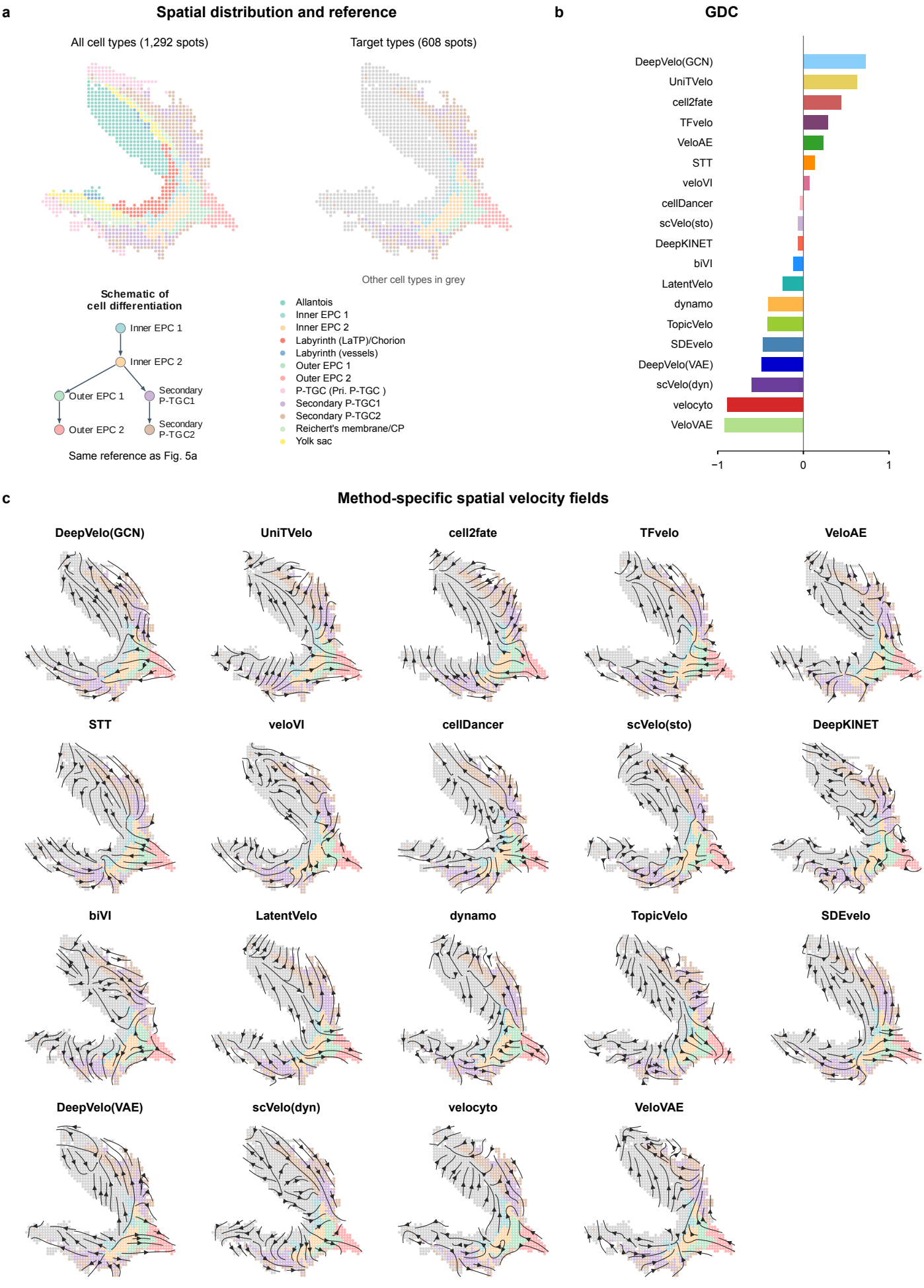

a
Spatial distribution and reference
All cell types (1,292 spots)
Target types (608 spots)
Other cell types in grey
Schematic of cell differentiation
Inner EPC 1
Inner EPC 2
Outer EPC 1
Secondary P-TGC1
Outer EPC 2
Secondary P-TGC2
Same reference as Fig. 5a
Allantois
Inner EPC 1
Inner EPC 2
Labyrinth (LaTP)/Chorion
Labyrinth (vessels)
Outer EPC 1
Outer EPC 2
P-TGC (Pri. P-TGC)
Secondary P-TGC1
Secondary P-TGC2
Reichert's membrane/CP
Yolk sac
b
GDC
DeepVelo(GCN)
UniTVelo
cell2fate
TFvelo
VeloAE
STT
veloVI
cellDancer
scVelo(sto)
DeepKINET
biVI
LatentVelo
dynamo
TopicVelo
SDEvelo
DeepVelo(VAE)
scVelo(dyn)
velocyto
VeloVAE
−1
0
1
c
Method-specific spatial velocity fields
DeepVelo(GCN)
UniTVelo
cell2fate
TFvelo
VeloAE
STT
veloVI
cellDancer
scVelo(sto)
DeepKINET
biVI
LatentVelo
dynamo
TopicVelo
SDEvelo
DeepVelo(VAE)
scVelo(dyn)
velocyto
VeloVAE

Extended Data Fig. 9 | Spatial transcriptomics: reference, scores and method fields

a, Two views of all 1,292 supplied ST_raw spots in identical fixed spatial coordinates: all 12 cell types in their original colors, and the six target cell types (608 spots) highlighted in the Fig. 5a colors with the other 684 spots in grey. The reference schematic is reused directly from Fig. 5a. Its five arrows indicate predefined evaluation relations, not measured spatial trajectories. b, GDC values displayed in descending order. Saved corrected eval2 GDC is supplemented with DeepVelo(VAE) re-evaluation from its saved velocity and displayed spatial projection (GDC = -0.493). The corrected eval2 weighting uses the 608 spots in the six reference types as denominator; three archived scores were reproduced within 1e-6 to verify this protocol. TFvelo retains its separately recorded evaluation with its original denominator and preprocessing; the bars are not a new uniform-protocol ranking. c, Nineteen available method fields arranged left to right and top to bottom in the same descending-GDC order as panel b, with identical spot coordinates and a shared grey background for non-target types. TopicVelo uses native stochastic transitions rerun from the saved topic fits after excluding genes without variable U/S signal and topics with fewer than two usable genes. Eleven topics are integrated for ST_raw and five for ST_small, replacing the historical last-topic-only configuration. Its high-dimensional direction is the transition-displacement proxy T @ Ms - Ms, and its spatial field is projected directly from the native transition matrix. GDC is -0.419 for ST_raw and 0.484 for ST_small. One ST_raw spot has an undefined direction and is excluded from directional scoring; the reference denominator remains 608 spots. Historical base-scVelo output is not used as TopicVelo. Existing directions for other methods are retained. All panels are combined on one page.